Blaise Delattre

JSPS Postdoctoral Fellow in Trustworthy AI

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Research Interests

Certified robustness and trustworthy AI; Lipschitz-constrained neural networks; randomized smoothing; stability and efficiency in deep learning; robust foundation and multimodal models.

Academic & Research Positions

2026–Present **JSPS Postdoctoral Fellow**, Institute of Science Tokyo, Tokyo, Japan.
Theory and methods for certified robustness of multimodal AI systems. Host: Assoc. Prof. Yang Cao.

2025–2026 **Post-doctoral Researcher**, Institute of Science Tokyo, Tokyo, Japan.
Provable robustness and trustworthiness of foundation models. Supervisor: Assoc. Prof. Yang Cao.

2022–2025 **Research Scientist (CIFRE)**, Foxstream, Lyon, France.
Robust deep learning for computer-vision applications (concurrent with PhD).

2021 **Research Intern**, Université Paris-Dauphine PSL, Paris, France.

2019–2020 **BNP Paribas**, Paris, France. Research Intern → Quantitative Analyst.

Education

2022–2025 **PhD in Computer Science**, Université Paris-Dauphine PSL, Paris, France.
Supervisors: Alexandre Allauzen, Quentin Barthélemy. Defended 20 June 2025.
Jury: Francis Bach (Chair), Rémi Gribonval, Audrey Repetti, Claire Boyer, Franck Mamalet.

2020–2021 **Master 2 MASH** (Mathematics, Learning and Data Science),
Université Paris-Dauphine & ENS Ulm PSL — joint PSL degree.

2016–2019 **MSc in Financial Engineering & Quantitative Finance**,
Ensimag Grenoble INP & IAE Grenoble, Grenoble, France.

Awards & Distinctions

2026 **JSPS Postdoctoral Fellowship for Research in Japan (Standard)**.
FY2026 second recruitment: 5.6% acceptance rate (125 selected from 2,231 applications).

2025 ICLR *oral* (top ∼1–2%): *Accelerated Training through Iterative Gradient Propagation Along the Residual Path.*

2023 ICLR *spotlight* (top ∼5%): *A Unified Algebraic Perspective on Lipschitz Neural Networks.*

2022 ICML *oral* (top ∼1–2%): *A Dynamical System Perspective for Lipschitz Neural Networks.*

Publications

* denotes equal contribution.

NeurIPS 2026 **B. Delattre***, H. Wu*, W.Y.B. Lim, Y. Cao. *Joint Certification for Attributed Graphs: Beyond Topology-Only Robustness.*

ICML 2026 **B. Delattre**, H. Wu, P. Caillon, W.Y.B. Lim, Y. Cao. *Certified Robustness under Heterogeneous Perturbations via Hybrid Randomized Smoothing.*

ICML 2026 E. Fagnou, P. Caillon, **B. Delattre**, A. Allauzen. *Trading Complexity for Expressivity Through Structured Generalized Linear Token Mixing.*

ICLR 2026 P. Caillon, E. Fagnou, **B. Delattre**, A. Allauzen. *Scaling Direct Feedback Learning with Theoretical Guarantees.*

ECAI 2025 P. Caillon, A. Colagrande, E. Fagnou, **B. Delattre**, A. Allauzen. *Forward Only Learning for Orthogonal Neural Networks of Any Depth.* Vol. 413, pp. 2041–2048.

AISTATS 2025 **B. Delattre**, P. Caillon, E. Fagnou, Q. Barthélemy, A. Allauzen. *Bridging the Theoretical Gap in Randomized Smoothing.*

ICLR 2025 E. Fagnou, P. Caillon, **B. Delattre**, A. Allauzen. *Accelerated Training through Iterative Gradient Propagation Along the Residual Path.* **(Oral)**

ACL 2024 E. Fagnou, P. Caillon, **B. Delattre**, A. Allauzen. *Chain and Causal Attention for Efficient Entity Tracking.*

ICLR 2024 **B. Delattre**, A. Araujo, Q. Barthélemy, A. Allauzen. *The Lipschitz–Variance–Margin Tradeoff for Enhanced Randomized Smoothing.*

ICML 2023 **B. Delattre**, Q. Barthélemy, A. Araujo, A. Allauzen. *Efficient Bound of Lipschitz Constant for Convolutional Layers by Gram Iteration.*

ICLR 2023 A. Araujo, A. Havens, **B. Delattre**, A. Allauzen, B. Hu. *A Unified Algebraic Perspective on Lipschitz Neural Networks.* **(Spotlight)**

ICML 2022 L. Meunier*, **B. Delattre***, A. Araujo*, A. Allauzen. *A Dynamical System Perspective for Lipschitz Neural Networks.* **(Oral)**

Research Funding & Fellowships

2026–Present **JSPS Postdoctoral Fellowship for Research in Japan (Standard)**, Japan Society for the Promotion of Science; Institute of Science Tokyo, Japan. Fellowship commenced September 2026.

2026 **MEXT SPReAD grant** (Supporting Pioneering Research through AI for 1,000 Discovery challenges), AI for Science research funding.

2025–2026 **DSAI Young Researcher Activity Support Program**, Data Science & Artificial Intelligence Research Group for Social Good, Institute of Science Tokyo, Japan (internal research grant for young researchers).

2022–2025 CIFRE doctoral funding (industry–academia), Foxstream / ANRT, France.

Research Supervision

Apr–Sep 2024 Primary advisor (with A. Allauzen and P. Caillon) of **Zakariae Moutaouakil** (M2, Université Paris Cité), LAMSADE internship on randomized-smoothing robustness certificates.

2025–Present Co-advisor (with Y. Cao) of **Hengyu Wu** (Master, Institute of Science Tokyo) at the TDSAI Lab, on certified robustness for multimodal foundation models and VLMs.

2026–Present Co-advisor (with Asst. Prof. W.Y.B. Lim) of **Foo Ping** (Master, NTU Singapore), on certified robustness for RAG retrieval.

Teaching

2023–2025 **Trustworthy Machine Learning in Practice** (Lecturer), Executive Master, Université Paris-Dauphine PSL.

2026 **Guest lecture** on *Adversarial Robustness in Machine Learning*, in Assoc. Prof. Yang Cao's database course (Master's level), Institute of Science Tokyo, Japan.

Invited Talks & Tutorials

2026	Conference talk: *Certified Robustness for Graph and Multimodal Models*, NDBS 2026, Shanghai, China.
Jul 2026	Invited talk, Yunnan International Workshop on Trustworthy Foundation Models (YTFM2026), Kunming, China, 16–19 July 2026 (NSFC–JSPS collaborative research workshop).
2025	Tutorial (accepted): *Robust Certificates for Neural Networks*, ADC 2025, Bali, Indonesia.
2025	Invited talk: *Toward Certified Robustness in Foundation Models*, KTFM Workshop, Kobe, Japan.
2025	Invited talk: *Lipschitz-Constrained Networks for Certified Robustness*, JSPS–NSFC Joint Research Workshop.
2024	Reading group: *Spectral Norm Estimation in Deep Learning*, ENS Lyon, France.
2023	Reading group: *The Lipschitz–Variance–Margin Tradeoff*, UIUC Adversarial Attacks RG (online).
2022–23	Conference / workshop talks: *A Dynamical System Perspective for Lipschitz Neural Networks*, ICML 2022 (Baltimore), SCAI 2023 (Paris).

Academic Service

Area Chair	ICLR 2027, ICML 2027.
Conference reviewer	NeurIPS (2025–2026), ICLR (2025–2026), ICML (2024–2026), AISTATS (2026), AAAI (2026), KDD AI4Science (2026), L4DC (2024).
Journal reviewer	Transactions on Machine Learning Research (TMLR), IEEE TAC, Neurocomputing (2026), IPSJ Trans. on Databases (TOD, 2026), ACML Journal Track (2024).
PC member	ICICS 2026 (International Conference on Information and Communications Security).
Chair	Poster & Encore Track Chair, ADMA 2025, Kyoto, Japan.

Technical Skills & Languages

Programming	Python, PyTorch.
Languages	French (native), English (fluent).

Referees

Prof. Alexandre Allauzen
PhD supervisor
ESPCI Paris PSL / Université Paris-Dauphine PSL
alexandre.allauzen@dauphine.psl.eu

Assoc. Prof. Yang Cao
Postdoctoral host
Institute of Science Tokyo
cao@c.titech.ac.jp

Asst. Prof. Wei Yang Bryan Lim
Collaborator
College of Computing & Data Science, NTU Singapore
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Dr. Quentin Barthélemy
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Dr. Alexandre Araujo
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